COMMERCIAL ITEM DESCRIPTION

EXCAVATOR, DENTAL (Carbon Steel, Black's Formula)

The General Services Administration has authorized the use of this Commercial Item Description.

Salient characteristics:

This commercial item description covers thin bladed, carbon steel, dental excavators suitable for use in dental operative procedures.

Classification. Dental excavators covered by this commercial item description shall be of the following type and shape, as specified in "NOTE" section.

Type I - Black's, Modified, Double Ended.

Shape 1 - Hatchet, No. 10, 11
Shape 2 - Discoid, No. 38, 39

Type II - Wedelstaedt's, Modified, Double Ended.

Shape 1 - Hatchet, No. 8, 9

Requirements. The excavators shall be in accordance with the applicable requirements specified herein.

Material. The excavators shall be fabricated from water hardening carbon steel. The steel shall meet the requirements of ASTM A686 for Grade A (regular hardening), Class W1 1.00 to 1.20 percent carbon, spheroidized annealed condition, steel for chemical composition, raw material hardness, macrostructure, surface decarburization, carbide form and distribution, hardness penetration and fracture grain size.

Hardness. Operative ends and shanks of the finished instrument shall have a Rockwell hardness of C56 to C62. Handles shall have a minimum Rockwell hardness of C20.

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Style, design, and dimensions. Excavators shall be one piece, double ended, thin blade design. Type I excavators shall be of Dr. Black's design modified by Dr. Ferrier. Type II excavators shall be Dr. Wedelstaedt's design modified by Dr. Ferrier. The style, design and dimensions of the excavators shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 24288 for the applicable type and shape of excavator.

The excavators shall conform to the applicable Black's formula, as specified in the drawing and as specified below. Excavators shall be suitable for use in hand excavating and cavity preparations, cleaving or splitting enamel and removal of dentine.

Instrument Number and Applicable Black's Formula.

Type	Shape	Instrument number	Black's Formula
I	1, Hatchet	10, 11	15-8-14
I	2, Discoid	38, 39	11 1/2-7-14
II	1, Hatchet	8, 9	10-7-14

Finish.

Operative ends. The excavator blades shall have sharp cutting edges and shall be free of defects such as coarse grind marks, cracks, burrs or feathers when examined under magnification, as specified herein.

Type I. The beveled portion shall have a sharp cutting edge. The operative ends (blades) including sharpened edges of the beveled portions shall have a fine honed finish.

Type II. The operative ends (blades) shall have a bright finish on the outer face and a fine honed finish on the inner face. The junction of the inner and outer faces (cutting edges) shall form a sharp edge around the periphery of the blade.

Handles and shanks. Excavator handles shall be straight and flat. Handles and shanks of excavators shall be plated as specified herein and shall have a bright finish. Handles shall have 40 ± 10 fillet or notch-type serrations per inch on seven faces and one smooth face for identification marking. Depth of serrations shall be 0.002 to 0.004 inch.

Plating. Handles and shanks shall be chromium (nickel over copper undercoat) or nickel over copper plated.

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Corrosion resistance. The plated portions (handles and shanks) of the excavators shall meet the boil test requirements of ASTM F1089 when tested as specified herein.

Identification marking. The excavators shall be permanently and legibly marked on the smooth face of the handle. The excavators shall be marked with the respective instrument number and Black's formula as specified herein. The respective instrument number shall be placed on the handle no less than 5 mm. nor more than 12 mm. from the junction of the handle and shank of the applicable operative end.

Workmanship. The operative end, shank, and handle of the excavator shall be free from defects such as coarse grind marks, cracks, or burrs which may detract from their appearance or impair their serviceability, proper functioning or intended use.

Unit. Each (EA). One excavator of the type specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection shall be one excavator. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, except for operative end dimensions which shall be conducted at inspection level S-1, AQL 2.5 (percent defective), and sampling for tests (when applicable) shall be conducted at inspection level S-2. For tests, acceptance number shall be zero (0) for all sample sizes.

Examination. Examination shall be as specified herein. Examination shall include, but shall not be limited to the following:

Style and design for the applicable type, and shape excavator not as specified.
Cutting edge of operative end rounded or not sharp.
Cutting edge not free from nicks, burrs, feathers, indentations, or other irregularities that affect cutting performance.
Operative end finish not as specified.
Operative end not free of corrosion.
Operative end dimensions not as specified.
Plating of handle and shank not free from flaking, cracks, or blisters.
Plating of handle and shank incomplete or not continuous.
Handle and shank finish not as specified.
Handle not straight and flat.
Identification marking not present, not complete, not properly located, or illegible.
Excavators not free from dirt, polishing compound, or foreign matter.
Handle serrations not uniform or not complete.
Handle serration dimensions not as specified.

Examination under magnification. The instruments shall be examined under four power (4X) magnification for the following defects:

Defects

Cutting edge of operative end rounded or not sharp.
Cutting edge not free from nicks, burrs, feathers, indentations, or other irregularities that affect cutting performance.
Operative end finish not as specified.

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Tests.

Hardness. Finished instrument hardness shall be determined in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Boil Test. The excavators shall be subjected to the boil test as specified in ASTM F1089.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6520-00-476-1504	EXCAVATOR, DENTAL, Black's Formula 11 1/2, 7, 14; No. 38 and 39; Carbon Steel
6520-00-476-1526	EXCAVATOR, DENTAL, Black's Formula, 15, 8, 14; No. 10 and 11; Carbon Steel
6520-00-476-1527	EXCAVATOR, DENTAL, Black's Formula 10, 14; No. 8 and 9; Carbon Steel

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

PROJECT NO. 6520-2040

LOCATION: NCR DISK #5

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53601	EXCAVATOR, DENTAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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